INCH-POUND
MIL-DTL-22200/2E
14 February 2025
SUPERSEDING
MIL-E-22200/2D
19 September 1986

DETAIL SPECIFICATION SHEET

ELECTRODES, WELDING, COVERED (AUSTENITIC CHROMIUM-NICKEL STEEL)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-22200.

1. SCOPE

1.1 Scope. This specification covers shielded metal arc welding covered electrodes, for manually depositing austenitic chromium-nickel steel weld metal.

1.2 Classification. Electrodes are furnished in the types, classes, and sizes shown in table I, as specified (see 6.2).

TABLE I. Types, classes, positions of welding, and sizes.

Type	Class	Positions of welding	Sizes (diameter, inch)
MIL-308-15 MIL-308-16	1	All	1/16, 5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
MIL-308L-15 MIL-308L-16	1	All	1/16, 5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
MIL-308HC-15 MIL-308HC-16	1	All	1/16, 5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
MIL-309-15 MIL-309-16	1	All	1/16, 5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
MIL-309L-15 MIL-309L-16	1	All	1/16, 5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
MIL-309Nb-15 MIL-309Nb-16	1	All	1/16, 5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 3439

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.

TABLE I. Types, classes, positions of welding, and sizes – Continued.

Type	Class	Positions of welding	Sizes (diameter, inch)
MIL-310-15 MIL-310-16	1	All	1/16, 5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
	3	Flat	5/16
MIL-312-15 MIL-312-16	1	All	5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
	3	Flat	5/16
MIL-316-15 MIL-316-16	1	All	1/16, 5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
MIL-316L-15 MIL-316L-16	1	All	1/16, 5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
MIL-317-15 MIL-317-16	1	All	1/16, 5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
MIL-318-15 MIL-318-16	1	All	3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
MIL-330-15 MIL-330-16	1	All	1/16, 5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
MIL-347-15 MIL-347-16	1	All	1/16, 5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
MIL-347HC-15 MIL-347HC-16	1	All	1/16, 5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
MIL-349-15 MIL-349-16	1	All	1/16, 5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4
MIL-16.8.2-15 MIL-16.8.2-16	1	All	1/16, 5/64, 3/32, 1/8, 5/32
	2	Horizontal fillets and flat	3/16, 1/4

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-22200 - Electrodes, Welding, Covered: General Specification for

(Copies of this document are available online at https://quicksearch.dla.mil/.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN WELDING SOCIETY (AWS)

AWS A4.2M	-	Standard Procedures for Calibrating Magnetic Instruments to Measure the Delta Ferrite Content of Austenitic and Duplex Ferritic-Austenitic Stainless Steel Weld Metal
AWS A5.4/A5.4M		Specification for Stainless Steel Electrodes for Shielded Metal Arc Welding

(Copies of these documents are available online at www.aws.org.)

ASTM INTERNATIONAL

ASTM A240/A240M	-	Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM A666	-	Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate, and Flat Bar

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Electrodes furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) before contract award (see 4.2 and 6.4).

3.2 Electrodes. Electrodes furnished under this specification shall be in accordance with the requirements of MIL-DTL-22200 and as specified herein.

3.3 Covering composition. The chemical composition of the electrode covering material is at the manufacturer's discretion, subject to the requirements of MIL-DTL-22200. Covering moisture shall be as specified in table VI.

3.4 Extent of covering at arc ends. Tubular electrodes at the arc end may extend beyond the coating in excess of the requirement in accordance with MIL-DTL-22200 provided the actual volume of the exposed electrode is not greater than the following:

$$1/32\ inch\ X\ \pi\left(\frac{Nominal\ diameter\ in\ inches}{2}\right)^2$$

3.5 Groove welds.

3.5.1 Mechanical properties of deposited weld metal. The mechanical properties of weld metal deposited in groove joints shall be as specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.

TABLE II. Mechanical properties of deposited weld metal.

Type	Ultimate tensile strength (psi) (minimum)	Elongation in 2 inches 1/ (percent) (minimum)	Bend test condition		Bend test requirement fissures per specimen 2/, 3/, 4/ (maximum)
			As-welded	Heat-treated	
MIL-308-15 and 16	80,000	35	X	---	3
MIL-308L-15 and 16	75,000	35	X	---	2
MIL-308HC-15 and 16	82,000	35	X	---	3
MIL-309-15 and 16	80,000	30	X	---	2
MIL-309L-15 and 16	75,000	30	X	---	2
MIL-309Nb-15 and 16	80,000	30	X	---	2
MIL-310-15 and 16	80,000	30	X	---	4
MIL-312-15 and 16	95,000	22	X	---	1
MIL-316-15 and 16	75,000	30	X	---	3
MIL-316L-15 and 16	70,000	30	X	---	3
MIL-317-15 and 16	80,000	30	X	---	3
MIL-318-15 and 16	80,000	25	X	---	2
MIL-330-15 and 16	75,000	25	X	---	4
MIL-347-15 and 16	75,000 as-welded, and heat-treated	30 as-welded, and 25 heat-treated	X	X	2
MIL-347HC-15 and 16	82,000 as-welded, and heat-treated	30 as-welded, and 25 heat-treated	X	X	3
MIL-349-15 and16	100,000	25	X	---	3
MIL-16.8.2-15 and 16	80,000 as-welded, and heat-treated	35 as-welded, and heat-treated	X	X	2

NOTES:

1/ Elongation not required for transverse weld specimens.

2/ Bend specimens shall not exhibit more than the specified number of fissures after completion of bending. The presence of a single tear or fissure having a dimension larger than 1/16 inch measured in any direction shall be cause for rejection or failure under test.

3/ Fissures 1/64 inch and less shall not be counted.

4/ The corners of specimens shall not be considered when evaluating the results.

3.5.2 Chemical composition of deposited weld metal. Chemical composition of deposited weld metal shall be as specified in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.

TABLE III. Chemical composition of deposited weld metal. [1/]

Type	Carbon	Manganese	Phosphorus	Sulfur	Silicon	Chromium	Nickel	Molybdenum	Niobium	Tungsten
	Percentage (single values are maximum)									
MIL-308-15 and 16 [2/]	0.08	0.5-2.5	0.04	0.03	0.90	18.0-21.0	9.0-11.0	---	---	---
MIL-308L-15 and 16 [2/]	0.04	0.5-2.5	0.04	0.03	0.90	18.0-21.0	9.0-11.0	---	---	---
MIL-308HC-15 and 16 [3/]	0.08-0.15	0.5-2.5	0.03	0.03	[4/]	18.0-21.0	9.0-11.0	---	---	---
MIL-309-15 and 16 [2/]	0.10	0.5-2.5	0.04	0.03	0.90	22.0-25.0	12.0-14.0	---	---	---
MIL-309L-15 and 16 [2/]	0.04	0.5-2.5	0.04	0.03	0.90	22.0-25.0	12.0-14.0	---	---	---
MIL-309Nb-15 and 16 [2/, 5/]	0.12	0.5-2.5	0.04	0.03	0.90	22.0-25.0	12.0-14.0	---	Niobium +Tantalum 0.70-1.00	---
MIL-310-15 and 16 [2/]	0.20	1.0-2.5	0.03	0.03	0.75	25.0-28.0	20.0-22.5	---	---	---
MIL-312-15 and 16 [2/]	0.15	0.5-2.5	0.04	0.03	0.90	28.0-32.0	8.0-10.5	---	---	---
MIL-316-15 and 16 [2/]	0.08	0.5-2.5	0.04	0.03	0.90	17.0-20.0	11.0-14.0	2.0-2.5	---	---
MIL-316L-15 and 16 [2/]	0.04	0.5-2.5	0.04	0.03	0.90	17.0-20.0	11.0-14.0	2.0-2.5	---	---
MIL-317-15 and 16 [2/]	0.08	0.5-2.5	0.04	0.03	0.90	18.0-21.0	12.0-14.0	3.0-4.0	---	---
MIL-318-15 and 16 [2/]	0.08	0.5-2.5	0.04	0.03	0.90	17.0-20.0	11.0-14.0	2.0-2.5	[6/]	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.

TABLE III. Chemical composition of deposited weld metal – Continued. [1/]

Type	Carbon	Manganese	Phosphorus	Sulfur	Silicon	Chromium	Nickel	Molybdenum	Niobium	Tungsten
	Percentage (single values are maximum)									
MIL-330-15 and 16 [2/]	0.25	1.0-2.5	0.04	0.03	0.90	14.0-17.0	33.0-37.0	---	---	---
MIL-347-15 and 16 [2/]	0.08	0.5-2.5	0.04	0.03	0.90	18.0-21.0	9.0-11.0	---	[7/]	---
MIL-347HC-15 and 16 [2/]	0.08-0.15	0.5-2.5	0.025	0.025	[4/]	18.0-21.0	9.0-11.0	---	[8/]	---
MIL-349-15 and 16	0.07-0.13	0.5-2.5	0.03	0.03	0.90	18.0-21.0	8.0-10.0	0.35-0.65	0.75-1.2 [9/]	1.25-1.75
MIL-16.8.2-15 and 16	0.10	0.5-2.5	0.03	0.03	0.50	14.5-16.5	7.5-9.5	1.0-2.0	---	---

NOTES:

1/ Analysis shall be made for the elements for which specific values are shown in this table. However, if the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine that the total of these other elements, except iron, is not present in excess of 0.70 percent.

2/ The ferrite number shall be from 4 to 10 as determined by an instrument calibrated in accordance with AWS A4.2M (see figure 1).

3/ The ferrite number shall not exceed 4 as determined by an instrument calibrated in accordance with AWS A4.2M (see figure 1).

4/ Silicon shall be not more than 5 times the actual carbon present.

5/ MIL-309Nb-15 and -16 was formerly named MIL-309Cb-15 and -16 (see 6.1.6).

6/ Niobium (combined Niobium+Tantalum) shall be not less than 6 times the actual carbon present but not more than 1.00 percent.

7/ Niobium (combined Niobium+Tantalum) shall be not less than 8 times the actual carbon present but not more than 1.00 percent. Tantalum shall not exceed 0.10 percent.

8/ Niobium (combined Niobium+Tantalum) shall be not less than 8 times the actual carbon present but not more than 1.4 percent.

9/ Titanium shall be limited to 0.15 percent maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.

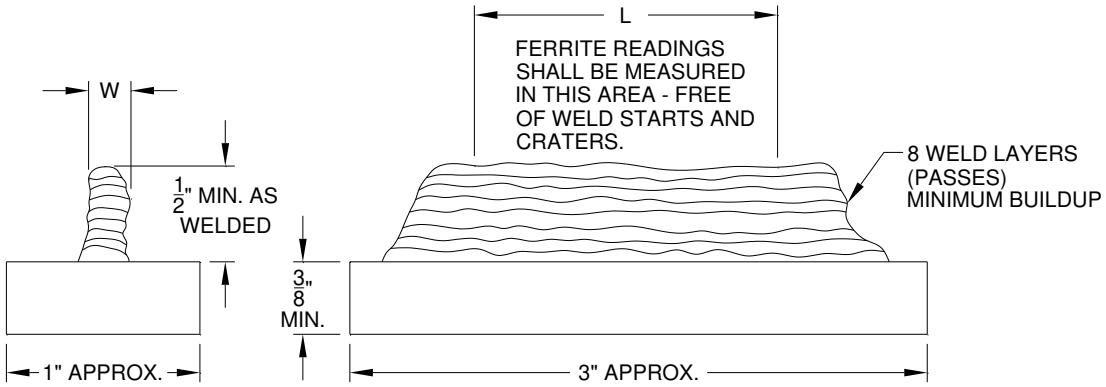


FIGURE 1. Ferrite weld specimen (see weld test number 4 of table VII).

NOTES:

1. Weld and plate dimensions shall be as specified on this figure and in table IV.
2. The base plate material for the weld pad shall be types 301, 302, or 304 in accordance with ASTM A240/A240M or ASTM A666.
3. The weld pad shall be built up by depositing single bead layers one on top of the other to a minimum height of ½ inch and a minimum of eight layers. The weld bead layers shall be deposited with a maximum weave width (electrode manipulation) of 2½ times the core wire diameter, except the first several layers may be deposited with a wider weave. The minimum width and approximate length of the weld buildup free of the weld arc starts and craters shall conform to table IV.
4. The welding current shall conform to table IV. The weld starts and stops shall be located at the ends of the weld buildup.
5. The weld pad shall be cooled between passes by water quenching immediately after each pass, except the last two weld passes shall be air cooled below 800 °F prior to water quenching.
6. Each weld pass shall be cleaned prior to depositing the next weld bead.
7. After welding, the weld buildup shall be surface ground to 64 root mean square finish to provide a finished surface for measuring the required ferrite content. The ground area need not be continuous along the specimen. The weld buildup shall be continuously water-cooled during the grinding operation. The finished ground surface shall be approximately parallel with the plate surface and shall be located within the last two passes or layers of weld buildup. Machining and metallographic polishing sufficient to remove the cold working produced by machining may be used in lieu of surface grinding. The weld buildup shall be water-cooled during the machining operation.
8. Ferrite readings shall be made in terms of ferrite numbers as determined by an instrument calibrated in accordance with AWS A4.2M. Instrument calibrating shall be verified not less than once per month and more frequently if necessary.
9. A total of six ferrite readings shall be taken on the finished surface along the longitudinal axis of the weld bead. The readings shall not be taken on either end where the weld craters (stop and start areas) are located. Readings shall be taken on rather flat areas having dimensions at least 5/32 inch wide. The six readings obtained shall be averaged to a single value for conversion to the ferrite number.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.

TABLE IV. Weld dimensions and welding parameters for ferrite weld specimen (see figure 1).

Electrode diameter (inch)	Width (W) minimum (inch)	Length (L) minimum (inch)	Welding current (amperes)	Arc voltage (volts)
1/16	1/4	3/4	35 to 45	1/
5/64	1/4	3/4	45 to 55	
3/32	1/4	1	65 to 80	
1/8	5/16	1	90 to 110	
5/32	5/16	1 1/2	120 to 140	
3/16	5/16	1 1/2	150 to 170	
1/4	5/16	1 1/2	180 to 210	

NOTE:
1/ The arc length shall be as short as practicable, and the voltage recorded for reference only.

3.6 Electrode lengths. The nominal lengths of electrodes shall be in accordance with table V.

TABLE V. Electrode lengths.

Sizes (inches)	Electrode lengths (inches)
1/16, 5/64	9
3/32	9 or 12 1/
1/8	9 or 14 1/
5/32 to 1/4 inclusive	14
5/16	18

NOTE:
1/ As specified (see 6.2).

3.7 Alloy identity of finished electrode. When specified (see 6.2), alloy identity of each finished covered electrode shall be in accordance with MIL-DTL-22200.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2 and table VI).

b. Conformance inspection (see 4.3 and table VI).

4.2 Qualification inspection. Qualification inspection shall include the examinations and tests of MIL-DTL-22200 and the additional tests in table VI herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.

TABLE VI. Required additional tests for qualification and conformance inspections. [1/]

Tests	Procedures	Requirements	Qualification	Conformance
Covering moisture	MIL-DTL-22200	[2/]	X	[3/]
Alloy identity – finished electrode	MIL-DTL-22200	3.7 herein	---	X [4/]
Weld tests	tables VII and VIII		table VII	table VIII

NOTES:

[1/] See 4.2 and 4.3 for additional requirements.

[2/] The covering moisture shall be determined as a qualifying test, and the moisture result obtained thereby shall be used to establish the maximum moisture level for the electrode type and diameter under this specification.

[3/] See 4.3.2.1.

[4/] Required when specified (see 3.7, 4.4, and 6.2).

4.3 Conformance inspection. Conformance inspection shall include the examinations and tests of MIL-DTL-22200 and the additional tests in table VI herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.

TABLE VII. Required weld tests for qualification. [1/]

Weld test assembly	Electrode type	Diameter sizes								Welding position	Tests					Test procedure	Requirements
		1/16	5/64	3/32	1/8	5/32	3/16	1/4	5/16		RT [2/]	Tensile	Bend	Chemical analysis [3/]	Ferrite content		
Test no. 3 (Figure 2)	All types	--	--	X	X	X	X	X	X	Flat	X	--	--	--	--	MIL-DTL-22200	MIL-DTL-22200
											--	X	X	--	--	MIL-DTL-22200	Table II herein
											--	--	--	X	--	MIL-DTL-22200 [4/]	Table III herein
Test no. 6 (Figure 3)	All types	X	X	--	--	--	--	--	--	Vertical	X	--	--	--	--	MIL-DTL-22200	MIL-DTL-22200
											--	X	--	--	--	MIL-DTL-22200	Table II herein
											--	--	--	X	--	MIL-DTL-22200 [4/]	Table III herein
Test no. 8a (Figure 4)	All types	--	--	X	X	X	--	--	--	Vertical	X	--	--	--	--	MIL-DTL-22200	MIL-DTL-22200
Test no. 4 (Figure 1)	MIL-308 MIL-308L MIL-308HC MIL-309 MIL-309L MIL-309Nb MIL-347	X	X	X	X	X	X	X	X	Flat	--	--	--	--	X	MIL-DTL-22200	Table III herein

NOTE:

1/ For qualification testing of the electrodes, all electrode types shall be tested using direct current electrode positive (DCEP). In addition, MIL-XXX-16 electrode types shall be tested using alternating current (AC). For conformance inspection, all electrode types shall be tested using DCEP.

2/ "RT" is Radiographic Testing.

3/ Instead of the weld pad specified in MIL-DTL-22200, chemical analysis samples may be taken from milling chips from ruptured ends of tensile specimens or from the ferrite test pad number (for those electrodes requiring ferrite determination).

4/ In case of chemical composition dispute, the referee method shall be as specified in AWS A5.4/A5.4M.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.

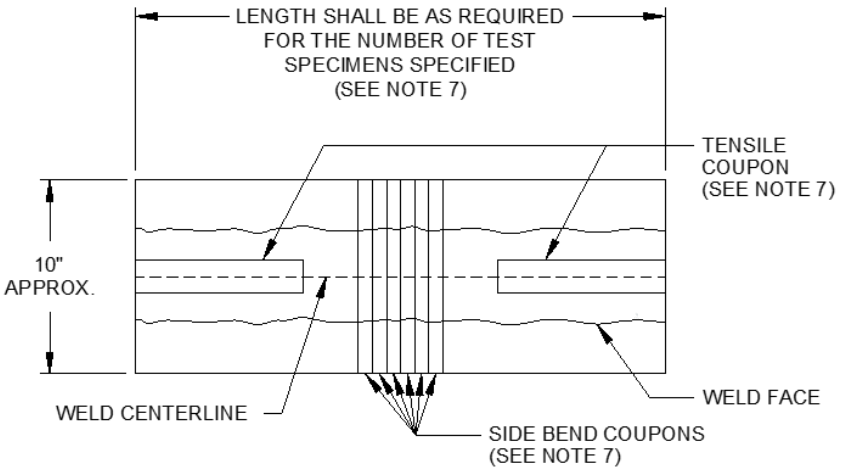
TABLE VIII. Required weld tests for conformance inspection.

Type	Required tests			Test procedures	Requirements
	Chemical analysis	Ferrite content	RT		
MIL-308	X	X	X	For chemical analysis: MIL-DTL-22200 1/	For chemical analysis: Table III herein
MIL-308L	X	X	X		
MIL-308HC	X	X	X		
MIL-309	X	X	X		
MIL-309L	X	X	X		
MIL-309Nb	X	X	X		
MIL-310	X	---	X	For ferrite content: Figure 1 herein	For ferrite content: Table III herein
MIL-312	X	---	X		
MIL-316	X	---	X		
MIL-316L	X	---	X		
MIL-317	X	---	X		
MIL-318	X	---	X		
MIL-330	X	---	X		
MIL-347	X	X	X	For RT: Figure 4 herein	For RT: MIL-DTL-22200
MIL-347HC	X	---	X		
MIL-349	X	---	X		
MIL-16.8.2	X	---	X		

NOTE:

1/ In case of chemical composition dispute, the referee method shall be as specified in AWS A5.4/A5.4M.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.



LAYOUT OF TEST PLATE

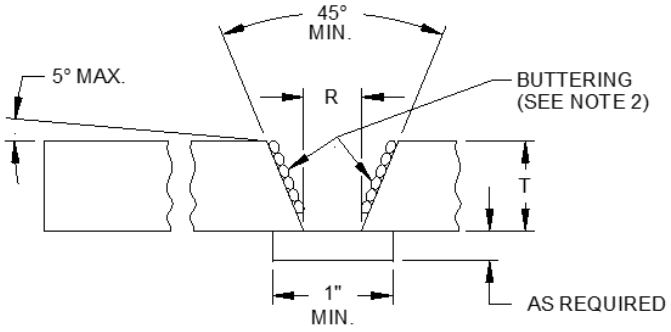


FIGURE 2. Weld joint for mechanical and bend tests (see weld test number 3 of table VII and table IX).

NOTES:

1. The base plate material shall match the chemistry of the electrode (same type designation in most instances). When base plate material matching the chemistry of the electrode is not available, American Iron and Steel Institute (AISI) type 306 and 347 material may be substituted for the fabrication of tension and bend specimens provided that for tension specimens buttering is accomplished. Base plate materials shall be in accordance with ASTM A240/A240M or ASTM A666.
2. If buttering is required, a minimum thickness of ⅛ inch shall be deposited on the plate edges comprising the groove. This may be done either prior to or after joint fit-up.
3. Preheat and interpass temperature shall not exceed 350 °F.
4. Welding shall be accomplished the flat position. Welding current shall be as specified in table IX. Welds shall be deposited in layers using the three-electrode core diameter maximum weave build up sequence. Layers shall be approximately ⅛ inch thick and shall be deposited in such a manner that the final layer includes a reinforcement of standard proportions (1/32 to ⅛ inch). Each layer shall be started at the finishing end of the preceding layer.
5. The weld reinforcement and backing strip shall be removed flush with base plate surfaces after completion of welding. The assembly shall be radiographed in accordance with MIL-DTL-22200. Radiographs shall meet the acceptance requirements of MIL-DTL-22200.
6. Tensile specimen from a ¾-inch test plate shall be 0.500 inch in diameter; those from a ½-inch test plate shall be 0.250 inch in diameter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.

FIGURE 2. Weld joint for mechanical and bend tests (see weld test number 3 of table VII and table IX) – Continued.

NOTES (continued):

7. The number and type of mechanical tests shall be as specified in table X.
8. Dimensions shall be as specified in table IX.
9. When heat treatment is required for electrode types MIL-347, MIL-347HC, and MIL-16.8.2 (see table X), the test coupons (prior to machining) shall be heated to 1,550 °F, held for 1 hour, and cooled. Heating and cooling rates shall not exceed 300 °F per hour.

TABLE IX. Groove preparation of test plate for test number 3 (see figure 2).

Electrode size (inch)	Minimum thickness (T) (inch)	Maximum root opening (R) (inch)	Number of layers 1/		Current (minimum)	
			Minimum	Maximum	DCEP (amperes)	AC (amperes)
3/32	1/2	1/4	1/	1/	75	85
1/8	1/2	1/4	1/	1/	95	105
5/32	3/4	1/2	7	9	135	150
3/16	3/4	1/2	7	9	155	170
1/4	3/4	1/2	7	9	245	265
5/16	3/4	1/2	7	9	345	365

NOTE:

1/ The pass and layer sequence shall be recorded.

TABLE X. Number of test specimens for mechanical and bend tests for number 3 weld test (see figure 2).

Electrode type	Size (inch)	Number of test specimens					
		Tensile 1/				Side bend 1/	
		0.250 inch		0.500 inch			
		As-welded	Heat-treated 2/	As-welded	Heat-treated 2/	As-welded	Heat-treated
MIL-347 2/ MIL-347HC 2/ MIL-16.8.2 2/	3/32 and 1/8	2	2	---	---	3	3
	5/32 to 5/16	---	---	2	2	3	3
All other	3/32 and 1/8	2	---	---	---	3	---
	5/32 to 5/16	---	---	2	---	3	---

NOTES:

1/ Tensile and side bend tests shall be conducted in accordance with MIL-DTL-22200.

2/ Heat treatment is required, see figure 2, note 9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.

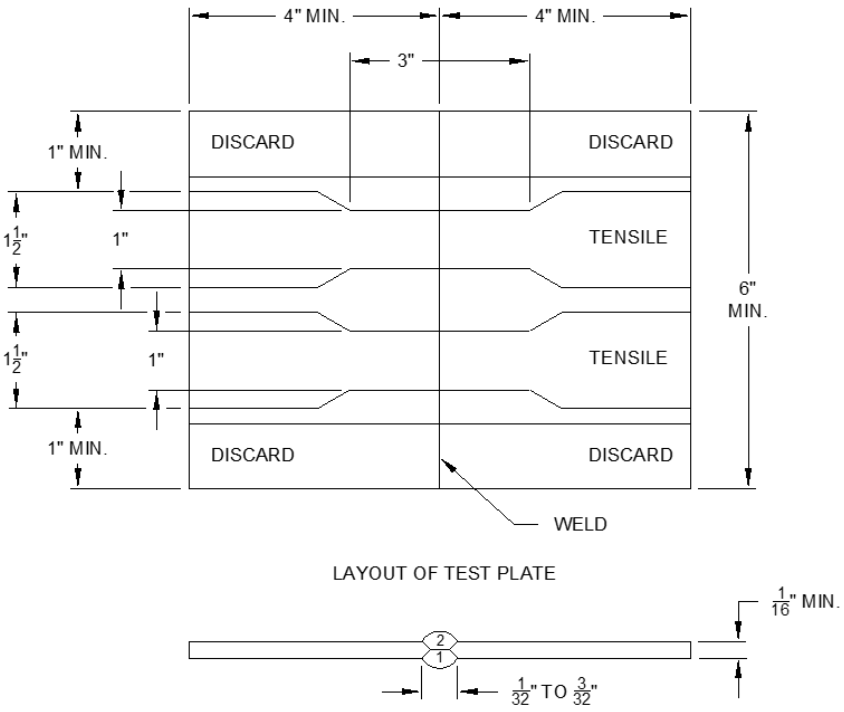


FIGURE 3. Weld joint preparation and layout of test for transverse weld tensile test for 1/16-inch and 5/64-inch electrode (see weld test number 6 of table VII).

NOTES:

1. Welding shall be done in the vertical position. Weld deposition shall be as shown. The backside of weld bead number 1 shall be cleaned by grinding prior to depositing weld bead number 2. Welding interpass temperature shall be 350 °F maximum.
2. The weld reinforcement shall be removed flush with the base plate on both surfaces in preparation for RT.
3. Two transverse weld tensile specimens shall be removed as shown and prepared and tested in the as-welded condition. Failure may be in weld or base plate material; however, to be acceptable, the strength shall be equal to or greater than that specified in table II for the type electrode being tested.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.

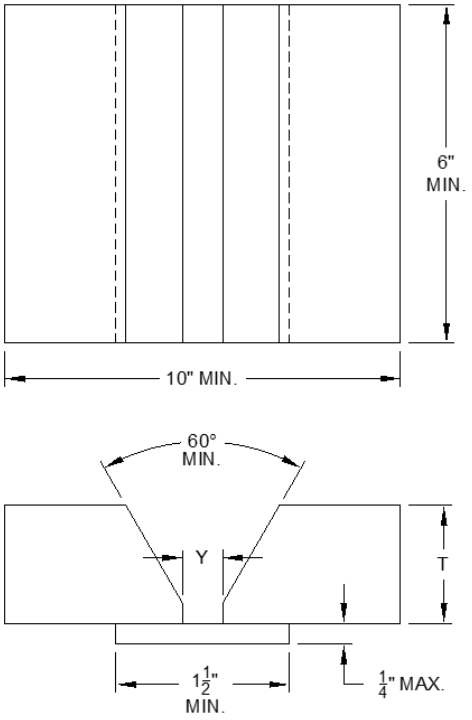


FIGURE 4. Groove weld usability test (see weld test number 8a of table VII).

NOTES:

1. Welding shall be accomplished with the test plate in the vertical or flat position as specified in table VII. Welding current shall be in accordance with the manufacturer's recommendations.
2. Base plate material shall be in accordance with AISI types 304, 316, 316L, or 347 in accordance with ASTM A240/A240M or ASTM A666.
3. The bevel edge preparation shall be machined or ground.
4. Welds shall be deposited in approximately ⅛-inch thick layers. Weld bead widths shall be as shown above. Each bead shall contain a start in the area to be evaluated.
5. The reinforcement shall be 3/16 inch maximum.
6. Preheat and interpass temperature shall be 60 °F minimum and 350 °F maximum.
7. After completion of the weld, the weld reinforcement and backing strip shall be removed flush with the base plate on both surfaces and the assembly shall be RT inspected in accordance with MIL-DTL-22200.
8. Plate thickness, root opening, and weld bead width shall be as specified in table XI.

TABLE XI. Dimensions for groove weld usability test (see figure 4).

Measurements	Electrode size		
	3/32 (inch)	⅛ (inch)	5/32 (inch)
Thickness (T) (minimum)	¼	⅜	½
Root opening (Y)	3/16 to 5/16	¼ to ⅜	5/16 to 7/16
Weld bead width (maximum)	⅜	½	⅝

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.

4.3.1 Lot. A lot of electrodes shall be in accordance with MIL-DTL-22200 for AWS classification C4.

4.3.2 Certification of conformance. A certificate of conformance (CoC) shall be in accordance with MIL-DTL-22200.

4.3.2.1 Covering moisture. The covering moisture content established during qualification for each electrode per table VI shall be reported on the CoC.

4.4 Alloy identity certification. When specified (see 6.2), the Government representative will affix their stamp and date on the container containing electrodes in accordance with MIL-DTL-22200. The letters "AT", to indicate alloy-tested, shall be affixed to the container adjacent to the date.

4.5 Inspection of packaging and marking. Inspection of packaging and marking shall be in accordance with MIL-DTL-22200.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The range of electrode types covered are several types of shielded metal arc welding covered electrodes of various chromium-nickel austenitic electrode types. This specification contains requirements above and beyond those of commercial specifications for use in critical military applications. Intended use of each type is as specified in 6.1.1 through 6.1.17 . Current stock of electrode types specified in this specification may be used until depleted; future requisitions for electrode types specified in this specification should refer to the current version of this specification.

6.1.1 MIL-308-15 and 16. This type is intended for the general welding of 18 percent chromium, 8 percent nickel, corrosion-resistant (stainless) steels.

6.1.2 MIL-308L-15 and 16. This type is intended for the general welding of 18 percent chromium, 8 percent nickel extra low carbon corrosion-resistant steel, used where intergranular corrosion should be limited but where corrosive conditions are not the most severe.

6.1.3 MIL-308HC-15 and 16. This type is intended for applications where low-magnetic permeability is of prime importance and where corrosion is not a factor.

6.1.4 MIL-309-15 and 16. This type is intended for the joining of the 18-8 grades of corrosion-resistant steel to mild steel; special treatment steel, class B armor, and other low-alloy air-hardening steels, when specified, and carbon and straight-chromium steels to austenitic grades. These electrodes are also intended for deposition of a corrosion-resistant overlay on carbon and low-alloy steels.

6.1.5 MIL-309L-15 and 16. This type is intended for similar applications as MIL-309-15 and 16 types but where added resistance to intergranular corrosion is desired.

6.1.6 MIL-309Nb-15 and -16. This type is intended for similar applications as the MIL-309-15 and 16 types, except for applications where expected corrosive conditions are more severe, or where higher strength is desirable at elevated temperature. MIL-309Nb-15 and -16 was formerly named MIL-309Cb-15 and -16. The change was made to conform to the worldwide uniform designation of the element niobium.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.

6.1.7 MIL-310-15 and 16. This type is intended for the welding of 25 percent chromium, 20 percent nickel heat-resistant alloy steels used in high-temperature service; special treatment steel, class B armor, and other low-alloy air-hardening steels, where specified; and carbon and straight-chromium steels to austenitic grades. These electrodes are also intended for deposition of a corrosion-resistant overlay on carbon and low-alloy steels.

6.1.8 MIL-312-15 and 16. This type is intended for joining super-alloys to low-alloy (chromium-molybdenum and chromium-nickel-molybdenum) steels.

6.1.9 MIL-316-15 and 16. This type is intended for the joining of corrosion-resistant alloys of similar designations. The molybdenum content imparts increased resistance to the corrosive effect of many organic acids, brine, sulfurous, and sulfuric acid.

6.1.10 MIL-316L-15 and 16. This type is intended for the welding of type 316L corrosion-resisting steel and under a condition similar to those under which type MIL-318 could be used.

6.1.11 MIL-317-15 and 16. This type is intended for welding of alloys of similar composition for increased corrosion-resistance to sulfuric and sulfurous acid and their salts.

6.1.12 MIL-318-15 and 16. This type is intended for the welding of type 318 corrosion-resisting steel. Generally, appropriate uses are similar to those of type MIL-316 electrodes except where elevated temperatures and corrosive conditions favor intergranular corrosion and where type alloy corrosion-resisting steel is used and the choice of a stabilized type MIL-318 electrode is indicated.

6.1.13 MIL-330-15 and 16. This type is used for repair welding of alloy castings and the joining of the cast, and wrought alloys of similar compositions, which are generally used for heat-resisting properties at high-temperatures.

6.1.14 MIL-347-15 and 16. This type is intended for the general welding of 18-8 corrosion-resisting steels stabilized by columbium, which are used where the condition of welding or service make unstabilized base metals susceptible to carbide precipitation with consequent intergranular corrosion. Use of these electrodes will not prevent intergranular corrosion alongside a weld in unstabilized 18-8 chromium-nickel base metals.

6.1.15 MIL-347HC-15 and 16. This type is intended for applications where low-magnetic permeability or heat resistance is of prime importance and where corrosion is not a factor.

6.1.16 MIL-349-15 and 16. This type is intended for the joints of the super-alloys for application requiring high strength at elevated temperature.

6.1.17 MIL-16.8.2-15 and 16. This type is intended for use in welding type MIL-316, MIL-317, and MIL-347 alloys for corrosion-resisting and high-temperature service in the as-welded or stress-relieved condition.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Type, class, and size required (see 1.2).

c. For 3/32-inch size electrode, whether 9- or 12-inch length is required (see 3.6 and table V).

d. For 1/8-inch size electrode, whether 9- or 14-inch length is required (see 3.6 and table V).

e. Whether alloy identity testing of the core wire of each finished covered electrode after final marking is required (see 3.7 and table VI, note 4).

f. Whether alloy identity testing and certification is required for each finished electrode (see 4.4).

g. Packaging requirements (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.

6.3 Inspection after delivery. Post-delivery inspection of electrodes to determine conformity to this specification and for acceptance thereof is the responsibility of the consignee. When receipt inspection testing types MIL-308, MIL-308L, MIL-309, MIL-309L, MIL-309Nb, and MIL-347 the consignee or user activity may consider the following delta ferrite test result limits acceptable:

a. A maximum ferrite number of 12.

b. An average minimum ferrite number of 3 with no individual inspection level of less than a ferrite number of 2.

Test results should meet the specified 4 to 10 ferrite number range. For type MIL-308HC, the user activity may consider a maximum ferrite number of 5 acceptable, while the contractor should meet the specified ferrite number of 4 as the maximum.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 22200 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.5 Subject term (key word) listing.

Alloy identity

Weld metal

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AR
Navy – SH
Air Force – 20

Preparing activity:
Navy – SH
(Project 3439-2023-003)

Review activities:
Navy – OS, YD
DLA – GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:21Z
Check the source to verify that this is the current version before use.